Source: Mohammed Sultan Farooqui on Unsplash

WOMEN IN THE LOOP

CARLY RICHMOND



AGENDA

- How incidents have changed
- HITL in incident management
- Benefits of AI
- AI challenges in incident management



Source: Mohammed Sultan Farooqui on Unsplash

ABOUT ME

- Senior Manager, Developer Advocacy @ elastic
- Frontend Engineer, Speaker & Blogger

SCAN ME





Source: Mehrpouya H on Unsplash

HOW WAS YOUR FIRST INCIDENT?

Source: Iván Díaz on Unsplash

2011



elastic

@CarlyLRichmond



Source: Sean Stratton on Unsplash

elastic

@CarlyLRichmond

Source: Iván Díaz on Unsplash

2018



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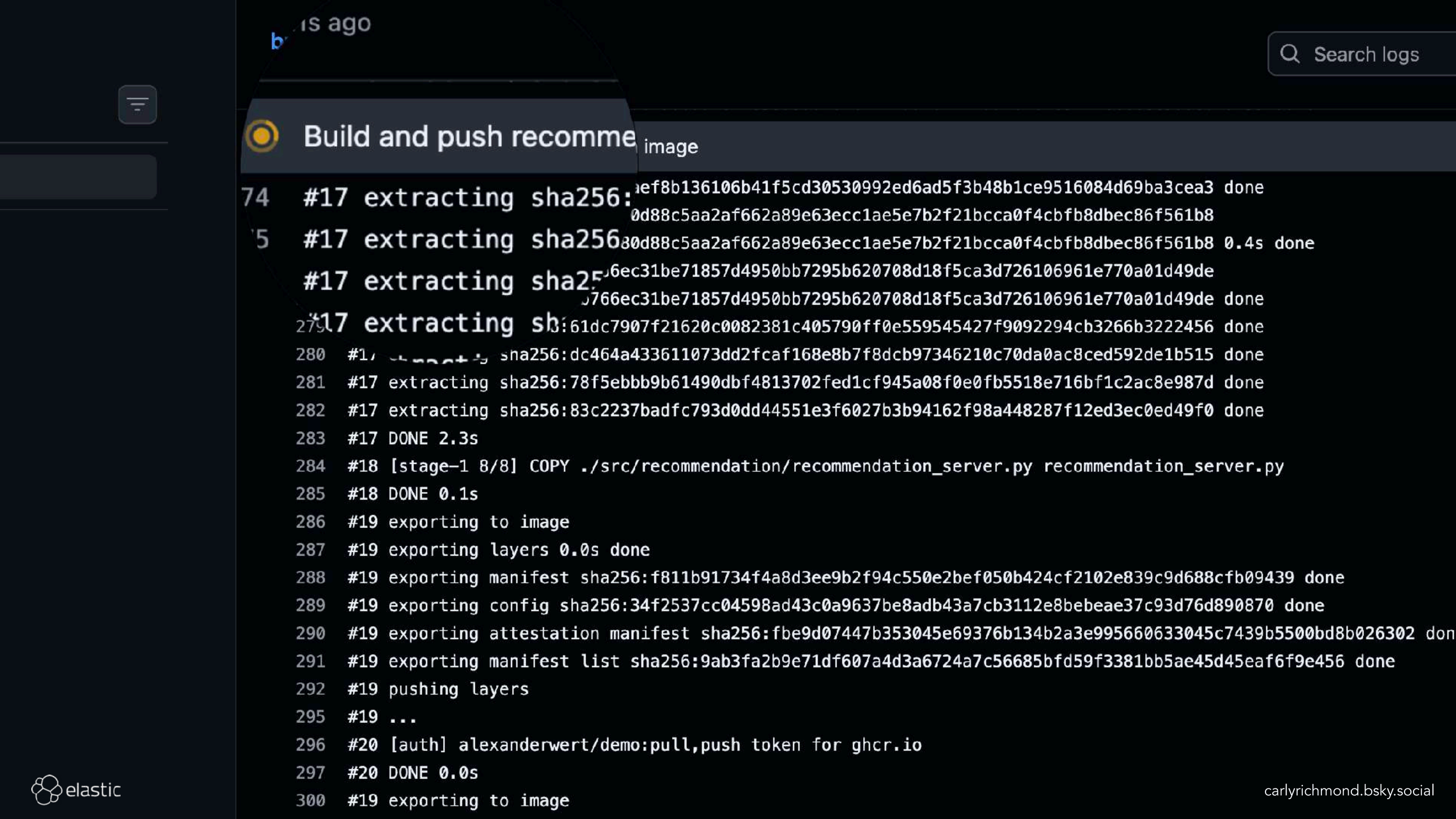


Source: Joran Quinten on Unsplash

Credit: Interrupted at work? You're not alone. | Caroline Fairchild @ Lean In, June 5 2026

"Men interrupt conversations about three times more often than women.
Women are twice as likely as men to report being interrupted or spoken over more than others."

–LEAN IN

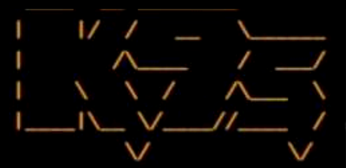

Search logs
Build and push recomme
image
#17 extracting sha256:
#17 extracting sha256
#17 extracting sha2
#17 extracting sh
#17 extracting sha256:dc464a433611073dd2fcaf168e8b7f8dcb97346210c70da0ac8ced592de1b515 done
#17 extracting sha256:78f5ebbb9b61490dbf4813702fed1cf945a08f0e0fb5518e716bf1c2ac8e987d done
#17 extracting sha256:83c2237badfc793d0dd44551e3f6027b3b94162f98a448287f12ed3ec0ed49f0 done
#17 DONE 2.3s
#18 [stage-1 8/8] COPY ./src/recommendation/recommendation_server.py recommendation_server.py
#18 DONE 0.1s
#19 exporting to image
#19 exporting layers 0.0s done
#19 exporting manifest sha256:f811b91734f4a8d3ee9b2f94c550e2bef050b424cf2102e839c9d688cfb09439 done
#19 exporting config sha256:34f2537cc04598ad43c0a9637be8adb43a7cb3112e8bebeae37c93d76d890870 done
#19 exporting attestation manifest sha256:fbe9d07447b353045e69376b134b2a3e995660633045c7439b5500bd8b026302 don
#19 exporting manifest list sha256:9ab3fa2b9e71df607a4d3a6724a7c56685bfd59f3381bb5ae45d45eaf6f9e456 done
#19 pushing layers
#19 ...
#20 [auth] alexanderwert/demo:pull,push token for ghcr.io
#20 DONE 0.0s
#19 exporting to image
elastic
carlyrichmond.bsky.social

IT WORKS?!

```
Cluster: gke_elastic-observability_europe-west3_alex-w-demo-otel-kubecon
User:    gke_elastic-observability_europe-west3_alex-w-demo-otel-kubecon
K9s Rev: v0.40.8 ⚡v0.50.18
K8s Rev: v1.34.3-gke.1444000
CPU:     41%
MEM:     61%
```

<1> default

<ctrl-d>	Delete	<l>	Logs	<f>	Show PortForward
<d>	Describe	<p>	Logs Previous	<t>	Transfer
<e>	Edit	<shift-f>	Port-Forward	<y>	YAML
<?>	Help	<z>	Sanitize		
<shift-j>	Jump Owner	<s>	Shell		



v1/pods(default)[25]

NAME↑	PF	READY	STATUS	RESTARTS	CPU	MEM	%CPU/R	%CPU/L	%MEM/R	%MEM/L	IP	NODE	AGE
accounting-bcccb46f-4b9lv	●	1/1	Running	0	4	147	n/a	n/a	98	98	10.48.3.215	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-qt85	147m
ad-5d9b7fcf7b-69d5b	●	1/1	Running	0	3	238	n/a	n/a	79	79	10.48.3.216	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-qt85	147m
cart-787df469f6-xh2rl	●	1/1	Running	0	4	74	n/a	n/a	46	46	10.48.3.217	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-qt85	147m
checkout-767b744d46-g7rlj	●	1/1	Running	1	2	15	n/a	n/a	77	77	10.48.3.218	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-qt85	147m
currency-f7fb8c49f-dcj2j	●	1/1	Running	0	2	7	n/a	n/a	39	39	10.48.3.219	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-qt85	147m
email-d6bf57b4d-62cqz	●	1/1	Running	0	2	66	n/a	n/a	66	66	10.48.3.220	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-qt85	147m
flagd-58ddc57565-rmsk9	●	2/2	Running	0	3	180	n/a	n/a	55	55	10.48.2.39	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	147m
fraud-detection-69fb6f45bb-jqzc6	●	1/1	Running	0	7	234	n/a	n/a	78	78	10.48.2.40	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	147m
frontend-dcb9588d9-spvxm	●	1/1	Running	0	25	120	n/a	n/a	48	48	10.48.3.221	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-qt85	147m
frontend-proxy-6fd5fb8d6d-62pcc	Ⓕ	1/1	Running	0	9	20	n/a	n/a	32	32	10.48.2.41	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	147m
image-provider-56b5b557cc-z7jpz	●	1/1	Running	0	1	12	n/a	n/a	24	24	10.48.0.187	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-2sfm	147m
kafka-8bcd9cdb8-mmvqc	●	1/1	Running	0	17	544	n/a	n/a	90	90	10.48.0.188	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-2sfm	147m
llm-c558f6b46-8n4t2	●	1/1	Running	0	21	70	n/a	n/a	n/a	n/a	10.48.3.222	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-qt85	147m
load-generator-69d5766587-z9xtb	●	1/1	Running	0	1549	1178	n/a	n/a	78	78	10.48.2.46	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	147m
otel-collector-agent-858j9	●	1/1	Running	0	34	103	n/a	n/a	51	51	10.48.3.183	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-qt85	23h
otel-collector-agent-fxk8k	●	1/1	Running	0	40	112	n/a	n/a	56	56	10.48.0.149	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-2sfm	23h
otel-collector-agent-pzsds	●	1/1	Running	0	74	110	n/a	n/a	55	55	10.48.2.244	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	23h
payment-57db774765-2ppjp	●	1/1	Running	0	23	96	n/a	n/a	69	69	10.48.2.42	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	147m
postgresql-6c9866bdf-9zg5f	●	1/1	Running	0	11	70	n/a	n/a	70	70	10.48.2.7	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	6d17h
product-catalog-84c5779894-ffpnv	●	1/1	Running	0	7	12	n/a	n/a	61	61	10.48.2.43	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	147m
product-reviews-7d6f59f467-49xs9	●	1/1	Running	0	23	73	n/a	n/a	73	73	10.48.2.44	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	147m
quote-bf89f65c4-vq286	●	1/1	Running	0	1	23	n/a	n/a	58	58	10.48.0.189	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-2sfm	147m
recommendation-9bbf4bf6f-jcmsd	●	1/1	Running	0	15	52	n/a	n/a	10	10	10.48.0.190	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-2sfm	147m
shipping-584c65cfb5-vd26b	●	1/1	Running	0	1	6	n/a	n/a	34	34	10.48.2.45	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	147m
valkey-cart-7784949c-jk2g5	●	1/1	Running	0	4	3	n/a	n/a	16	16	10.48.2.14	gke-alex-w-demo-otel-kube-custom-pool-6d24045d-h3o9	6d17h

Observability
Discover
Dashboards
Workflows
Alerts
Cases
SLOs
More

Add data

Alerts

Rule count 16 | Disabled 3 | Snoozed 0 | Errors 3

Manage Rules

Search alerts (e.g. kibana.alert.evaluation.threshold > 75)

Last 24 hours | Refresh

Status: active, recovered 2 | Rule | Group | Tags

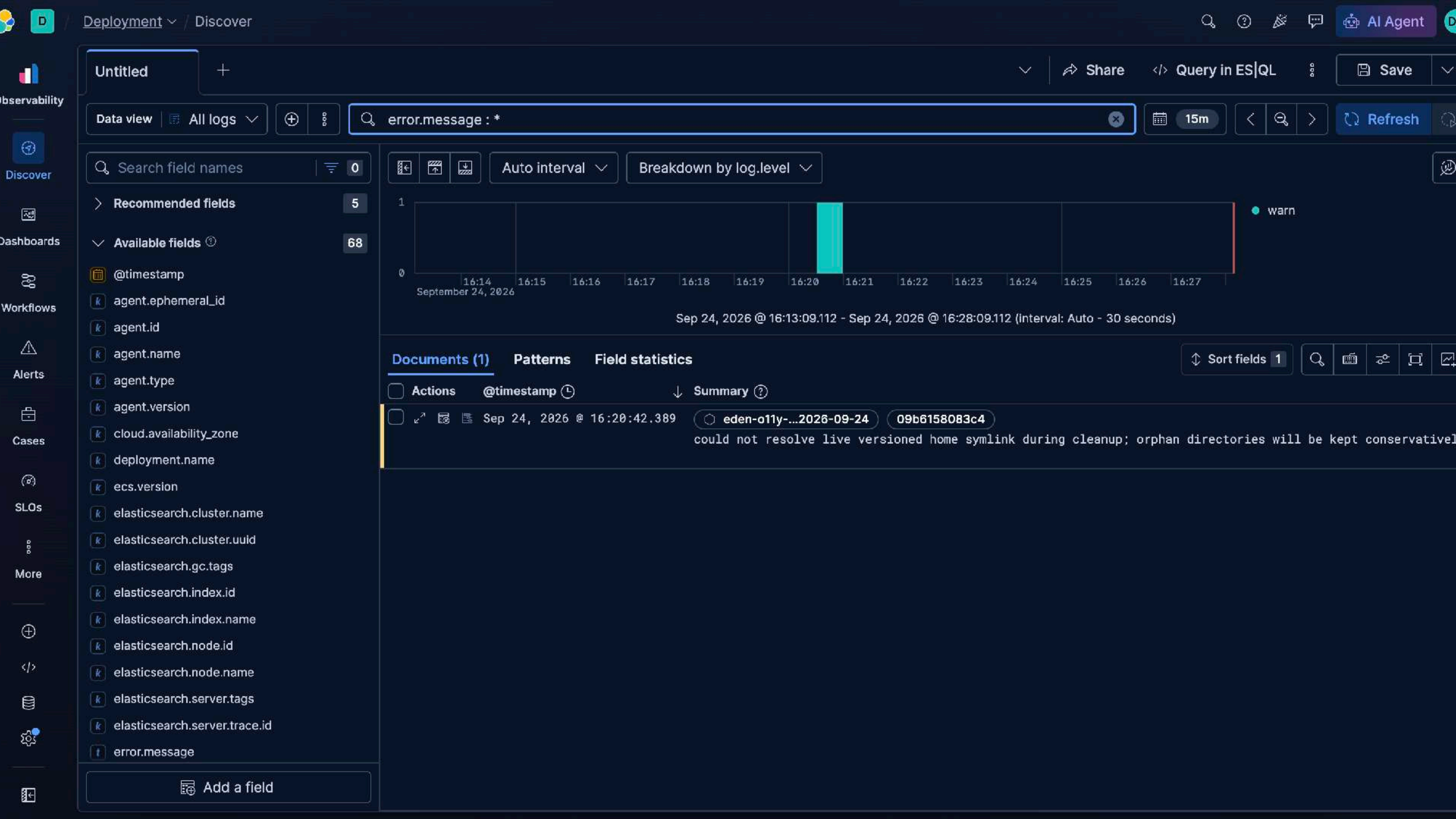
53 Alerts **0** Active now



Columns 10 | Sort fields 1 | 53 alerts | Fields

Updated 10 seconds ago | Group alerts by: None

Actions	Alert Status	Triggered	Duration	Rule name	Group	Observed val...	Threshold	Tags	Workflow tags	Reason
	Recovered	Sep 25, 2026, 13:47:06.040 (U	240 s	shares_traded_anomalies	shares_trade...	--	--	trading	--	No anomalie
	Recovered	Sep 25, 2026, 13:42:06.028 (U	60 s	shares_traded_anomalies	shares_trade...	--	--	trading	--	No anomalie
	Recovered	Sep 25, 2026, 13:33:05.993 (U	60 s	shares_traded_anomalies	shares_trade...	--	--	trading	--	No anomalie
	Recovered	Sep 25, 2026, 12:53:04.428 (U	60 s	shares_traded_anomalies	shares_trade...	--	--	trading	--	No anomalie
	Recovered	Sep 25, 2026, 12:22:04.379 (U	60 s	shares_traded_anomalies	shares_trade...	--	--	trading	--	No anomalie
	Recovered	Sep 25, 2026, 11:53:04.314 (U	240 s	shares_traded_anomalies	shares_trade...	--	--	trading	--	No anomalie


Deployment
Discover
AI Agent
Untitled
Share
Query in ES|QL
Save
Data view
All logs
error.message : *
15m
Refresh
Search field names
Recommended fields 5
Available fields 68
@timestamp
agent.ephemeral_id
agent.id
agent.name
agent.type
agent.version
cloud.availability_zone
deployment.name
ecs.version
elasticsearch.cluster.name
elasticsearch.cluster.uuid
elasticsearch.gc.tags
elasticsearch.index.id
elasticsearch.index.name
elasticsearch.node.id
elasticsearch.node.name
elasticsearch.server.tags
elasticsearch.server.trace.id
error.message
Add a field
Auto interval
Breakdown by log.level
warn
September 24, 2026
Sep 24, 2026 @ 16:13:09.112 - Sep 24, 2026 @ 16:28:09.112 (Interval: Auto - 30 seconds)
Documents (1)
Patterns
Field statistics
Sort fields 1
Actions
@timestamp
Summary
Sep 24, 2026 @ 16:20:42.389
eden-o11y-...2026-09-24
09b6158083c4
could not resolve live versioned home symlink during cleanup; orphan directories will be kept conservativel
Observability
Discover
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Cases
SLOs
More

Clusters

1	1	0
Total clusters	Healthy	Unhealthy

Cluster breakdown by health



Top 10 clusters by CPU utlization



	Avg	Min	Max
prod-3-o11y-superde...	7.97%	7.42%	8.55%

Top 10 clusters by memory working set utilization



	Avg	Min	M
prod-3-o11y-superde...	39.03%	38.94%	39.1

Top 10 clusters by disk utilization

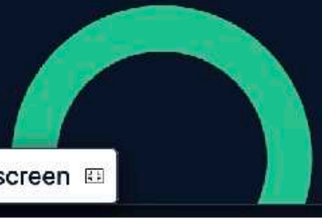


	Avg	Min	
prod-3-o11y-superde...	39.03GB	39.01GB	39.

Nodes

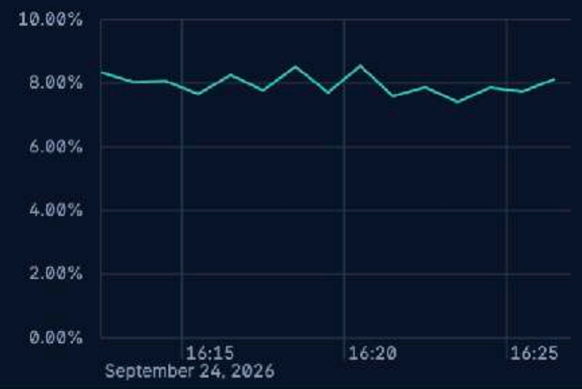
1	1	0
Total nodes	Ready	Not ready

Node breakdown by readiness



Exit full screen

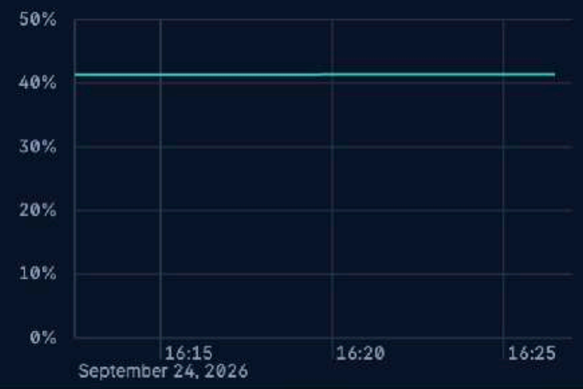
Top 10 nodes by CPU utlization



Top 10 nodes by memory working set utilization

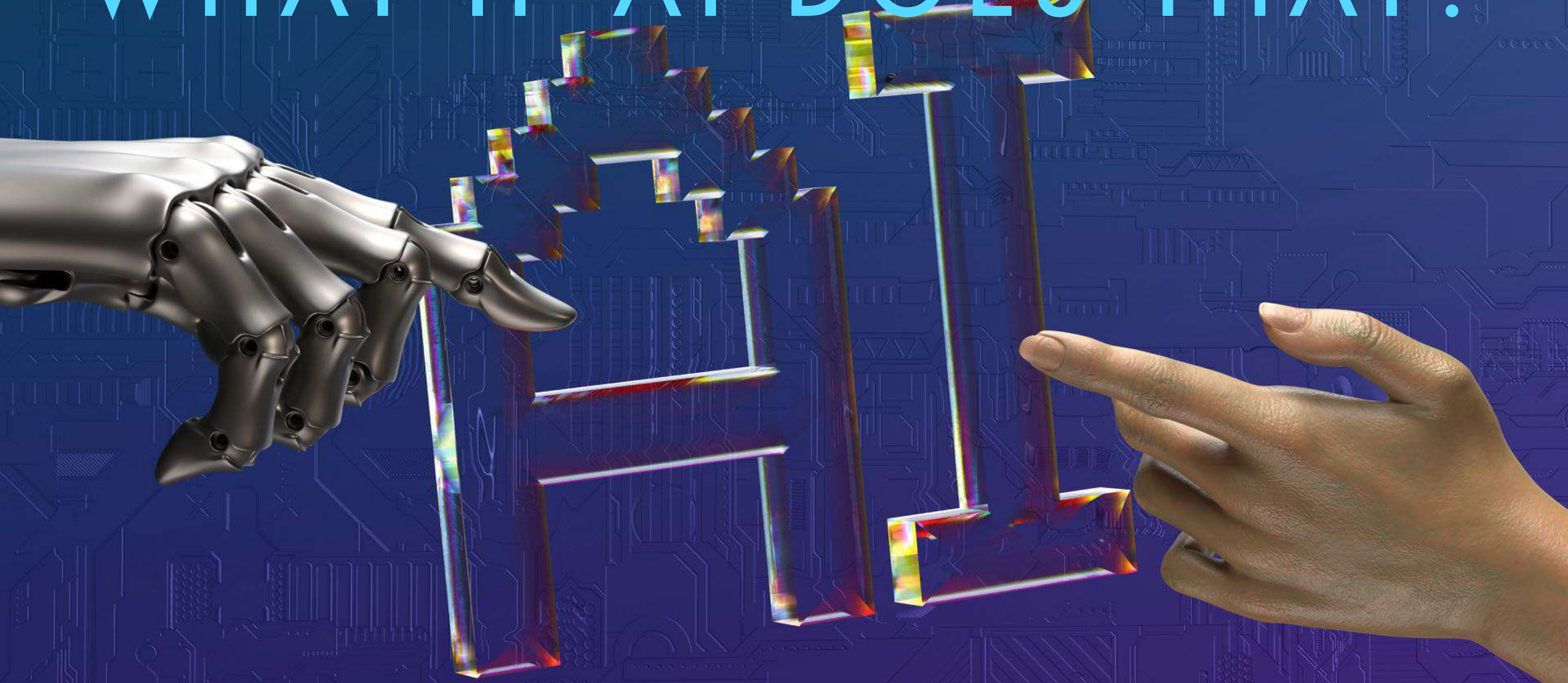


Top 10 nodes by disk utilization



Source: Igor Omilaev on Unsplash

WHAT IF AI DOES THAT?



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AI SRE AGENT · OPERATING MODEL

The agent investigates. The human decides.

An AI SRE agent can compress detection, correlation and diagnosis to seconds. Every action that touches production still passes a **human gate**, and every human decision is written back as signal the agent learns from.

- **Agent** — autonomous, read-only
- **Human** — judgement & authority
- **Gated** — write to production

01 Signal — MACHINE

- **SLO burn & alerts**
  Error budget burn rate, paging alerts, anomaly detection.
- **Telemetry**
  Logs, metrics, traces, change events and deploy history.
- **Human report**
  An on-call engineer asks the agent to look into something.

02 Agent investigates — NO WRITES

01 **Gather context**
Pull dashboards, recent deploys, related incidents and runbooks.

02 **Correlate & hypothesise**
Rank candidate causes with the evidence behind each one.

03 **Draft a remediation plan**
Concrete steps, blast radius, rollback path and confidence score.

03 Human gate — HUMAN

- **On-call SRE reviews the plan**
  Evidence, diff and rollback shown side by side.
- **Approve**
  Agent executes the plan as written, step by step.
- **Edit & approve**
  Human narrows scope or swaps a step, then releases it.
- **Reject & take over**
  Human states why, and drives the fix manually.

04 Act & verify — SUPERVISED

01 **Execute in scope**
Only the approved steps run, inside the approved blast radius.

02 **Verify recovery**
Watch the SLO and golden signals; auto-roll back if they worsen.

03 **Report back**
Timeline, actions taken and a draft incident write-up for review.

← FEEDBACK LOOP Every approval, edit and rejection is captured as **labelled judgement** — it tunes the agent's hypotheses, updates runbooks, and moves well-understood actions up an autonomy tier.

TIER 1 · AUTONOMOUS

Read and diagnose

Queries, correlation, summaries, draft plans. **No approval needed** — nothing changes in production.

TIER 2 · NOTIFY

Low-risk, reversible

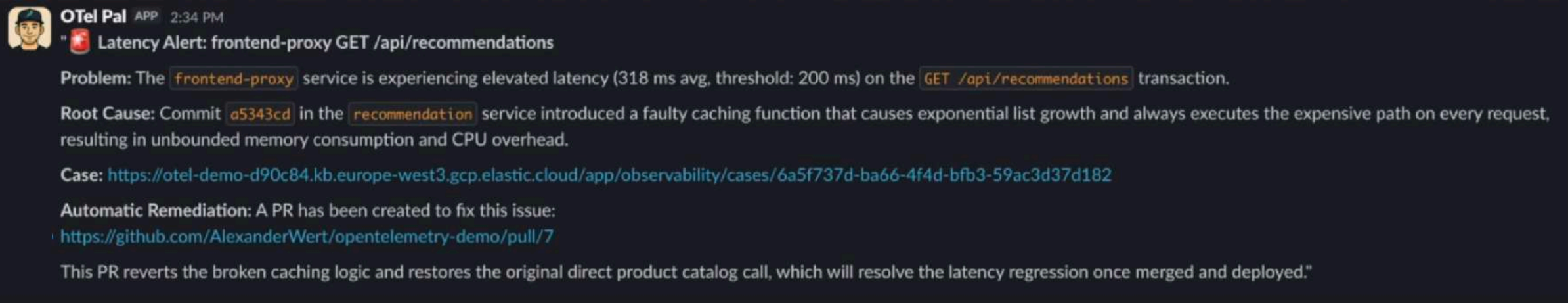
Scale a replica, clear a cache, silence a duplicate alert. **Runs, then tells you** — with one-click undo.

TIER 3 · APPROVAL REQUIRED

Irreversible or wide blast radius

Rollbacks, failovers, data or config changes, anything customer-facing. **A human approves first**, always.

EXAMPLE


OTel Pal APP 2:34 PM
" Latency Alert: frontend-proxy GET /api/recommendations
Problem: The frontend-proxy service is experiencing elevated latency (318 ms avg, threshold: 200 ms) on the GET /api/recommendations transaction.
Root Cause: Commit a5343cd in the recommendation service introduced a faulty caching function that causes exponential list growth and always executes the expensive path on every request, resulting in unbounded memory consumption and CPU overhead.
Case: https://otel-demo-d90c84.kb.europe-west3.gcp.elastic.cloud/app/observability/cases/6a5f737d-ba66-4f4d-bfb3-59ac3d37d182
Automatic Remediation: A PR has been created to fix this issue:
https://github.com/AlexanderWert/opentelemetry-demo/pull/7
This PR reverts the broken caching logic and restores the original direct product catalog call, which will resolve the latency regression once merged and deployed."

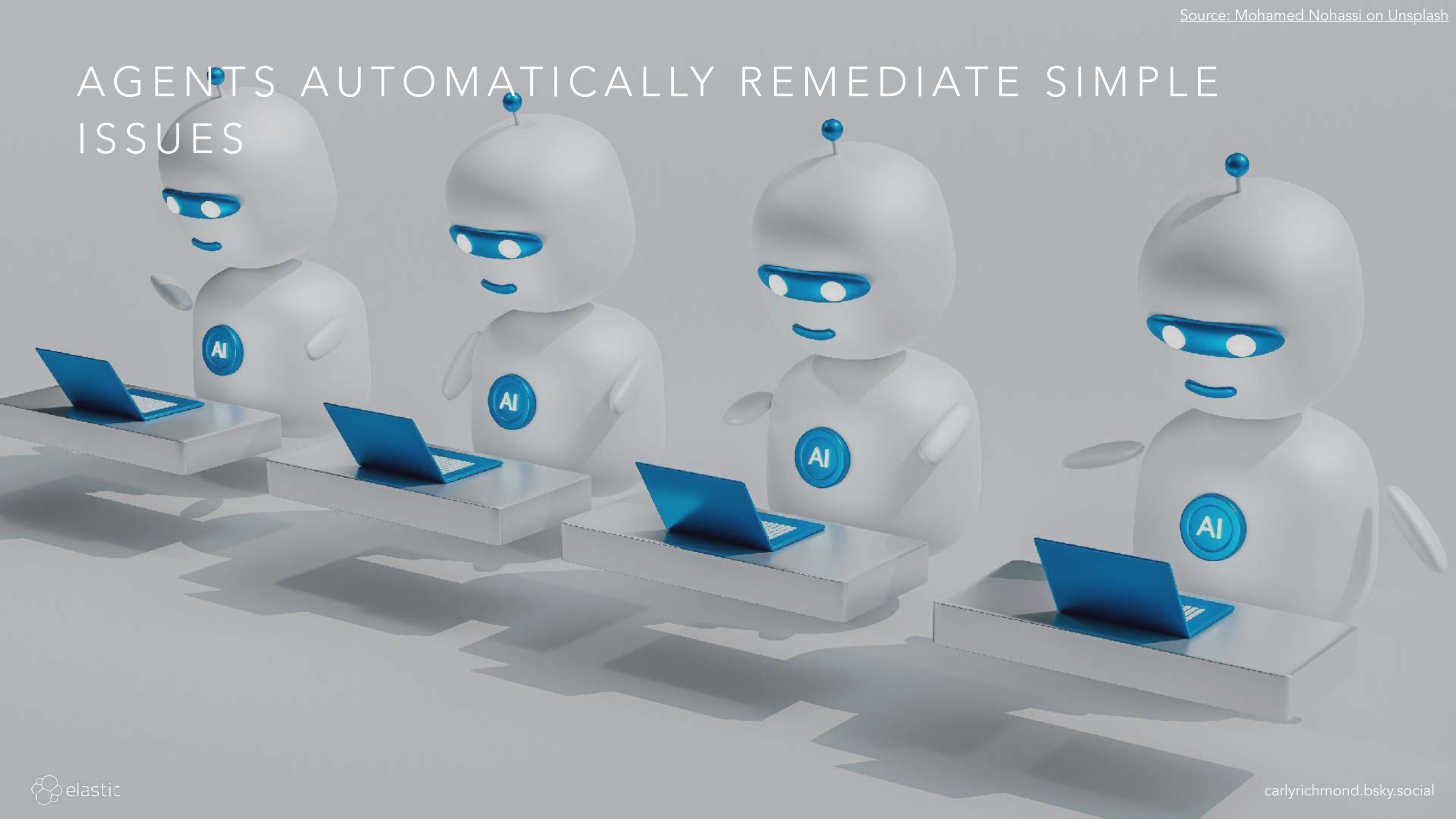


Source: Wai Siew on Unsplash

DO AI SRE AGENTS ELIMINATE THE INCIDENT BRIDGE?

Source: Mohamed Nohassi on Unsplash

AGENTS AUTOMATICALLY REMEDIATE SIMPLE ISSUES











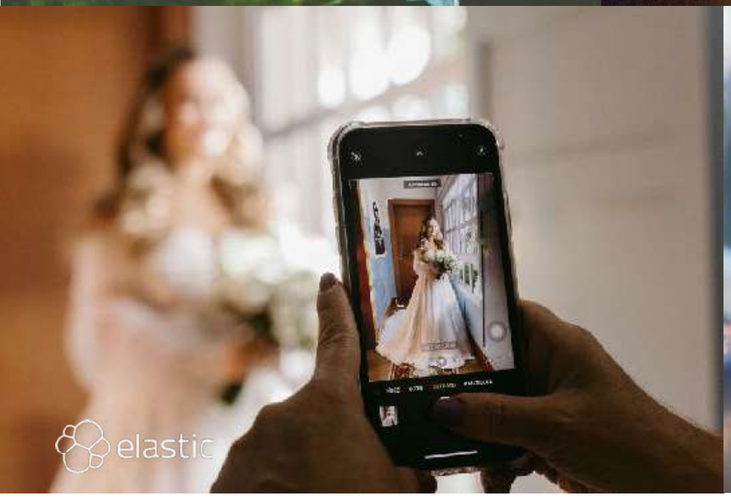


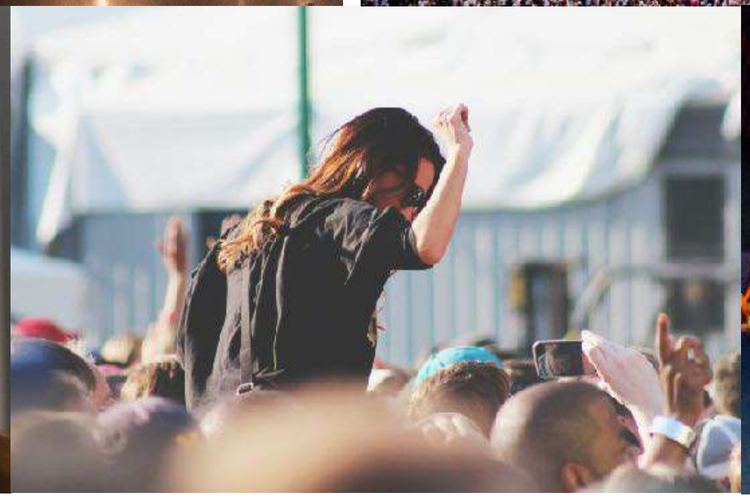
















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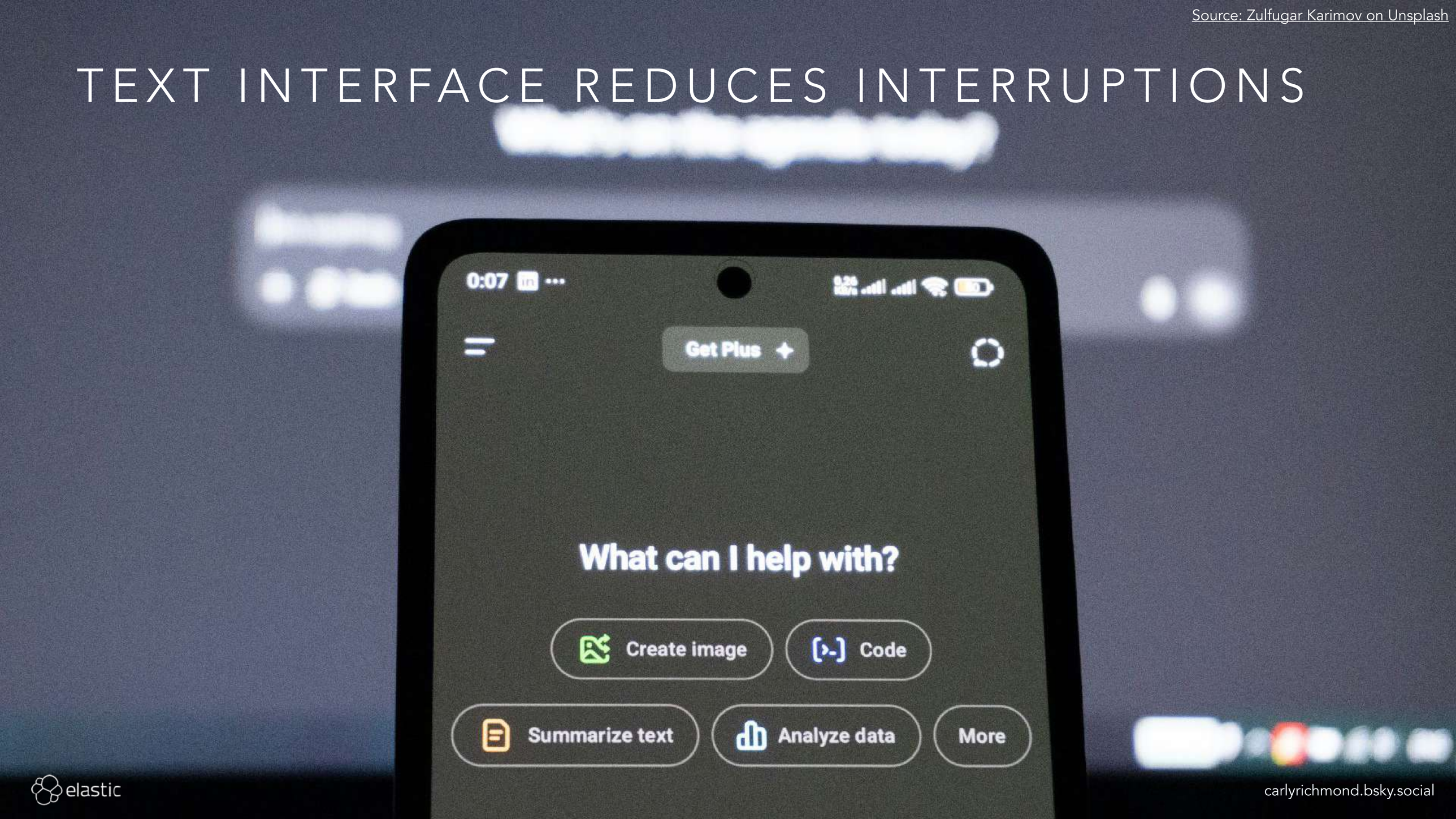
FLATTEN TRIBAL KNOWLEDGE



Source: Gabriella Clare Marino on Unsplash

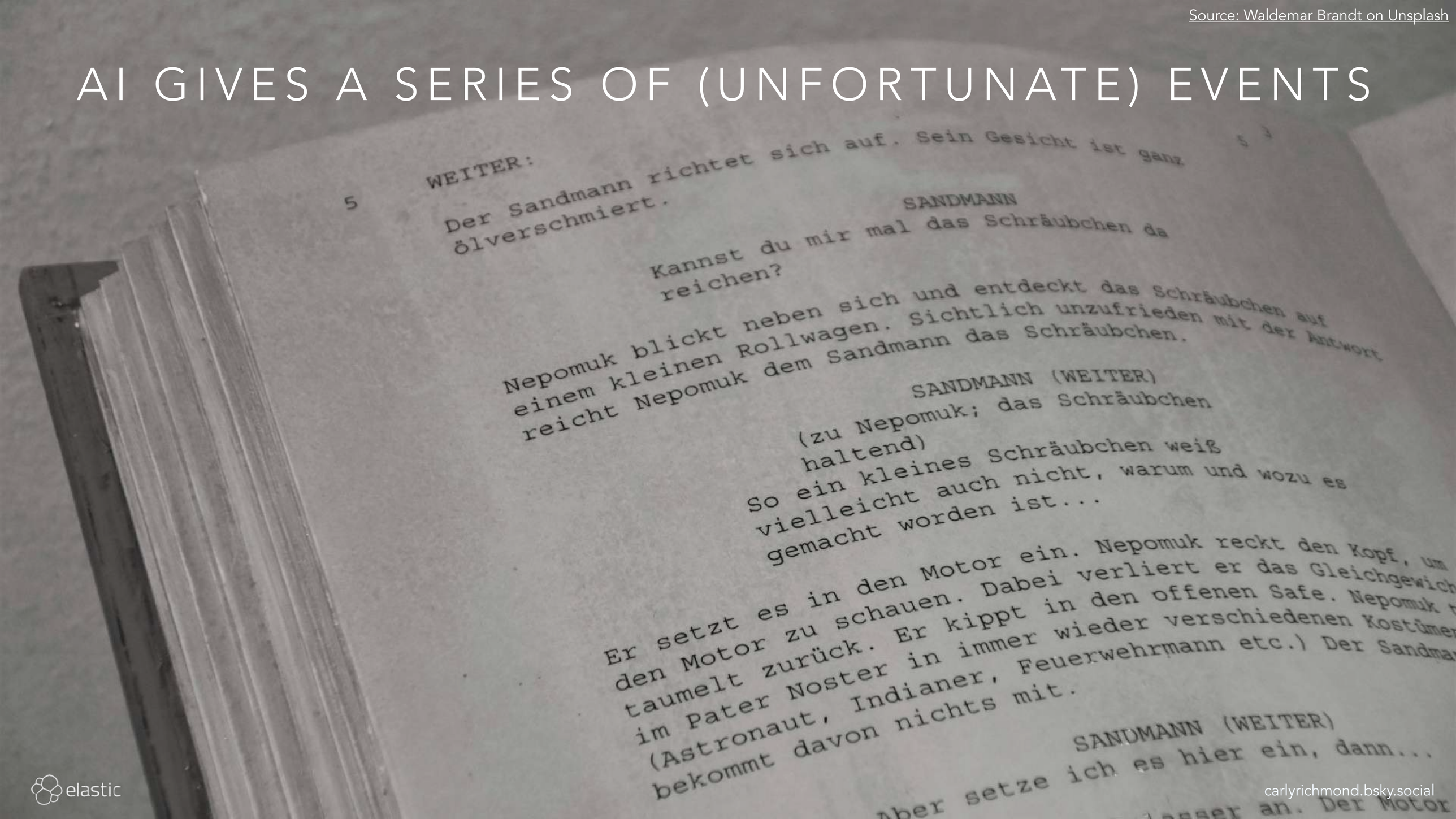
Source: Zulfugar Karimov on Unsplash

TEXT INTERFACE REDUCES INTERRUPTIONS



Source: Waldemar Brandt on Unsplash

AI GIVES A SERIES OF (UNFORTUNATE) EVENTS



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Source: Wai Siew on Unsplash

CHALLENGES OF AI IN INCIDENT RESPONSE

elastic

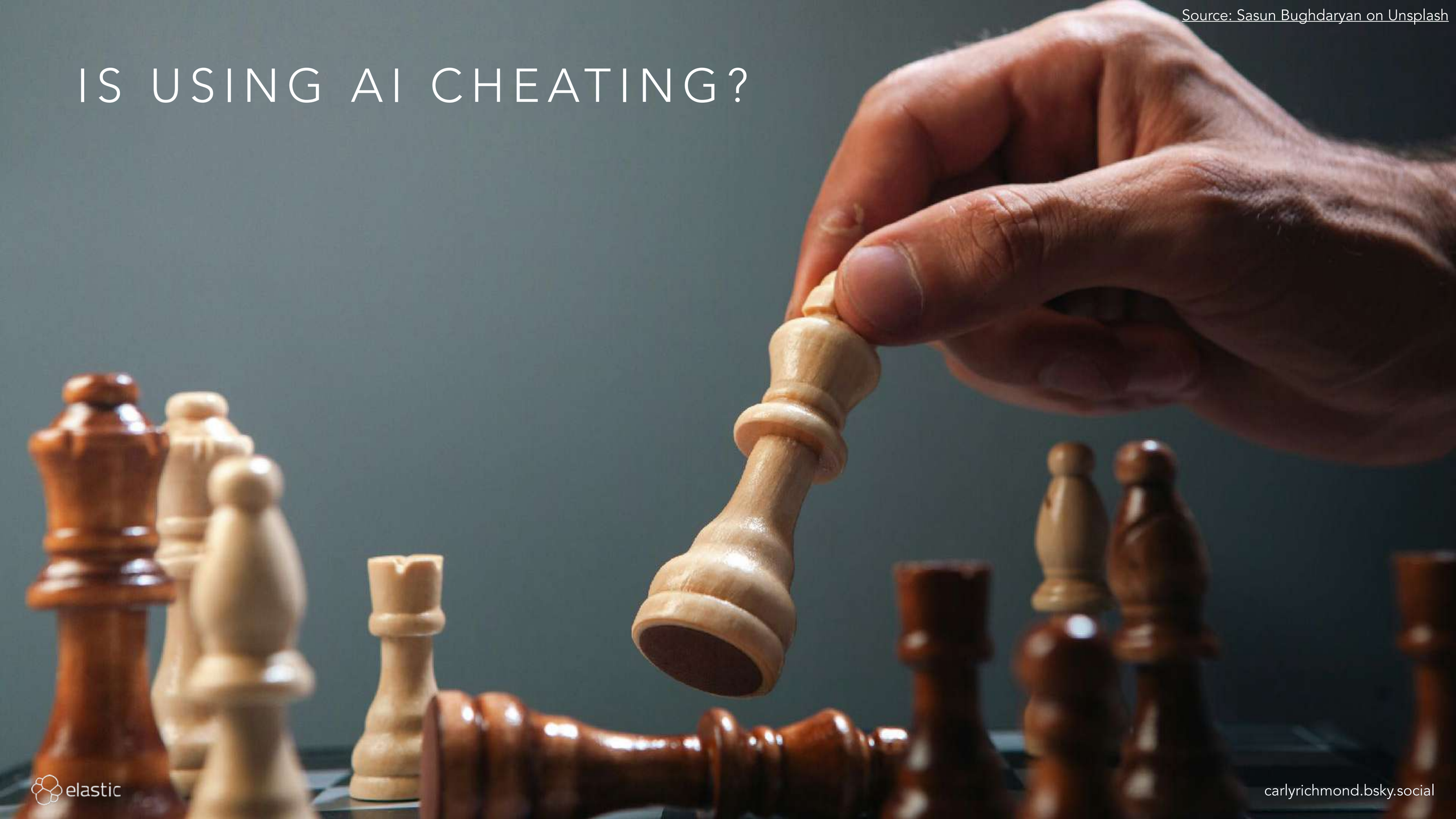
carlyrichmond.bsky.social

Credit: New research: Women use AI less often at work and get less credit | Mary Noble-Tolla @ Lean In

"Men are 22% more likely to use AI daily or constantly at their jobs (33% of men vs. 27% of women).
Men are 27% more likely to have been praised for doing so (23% of men vs. 18% of women)."

–LEAN IN

IS USING AI CHEATING?



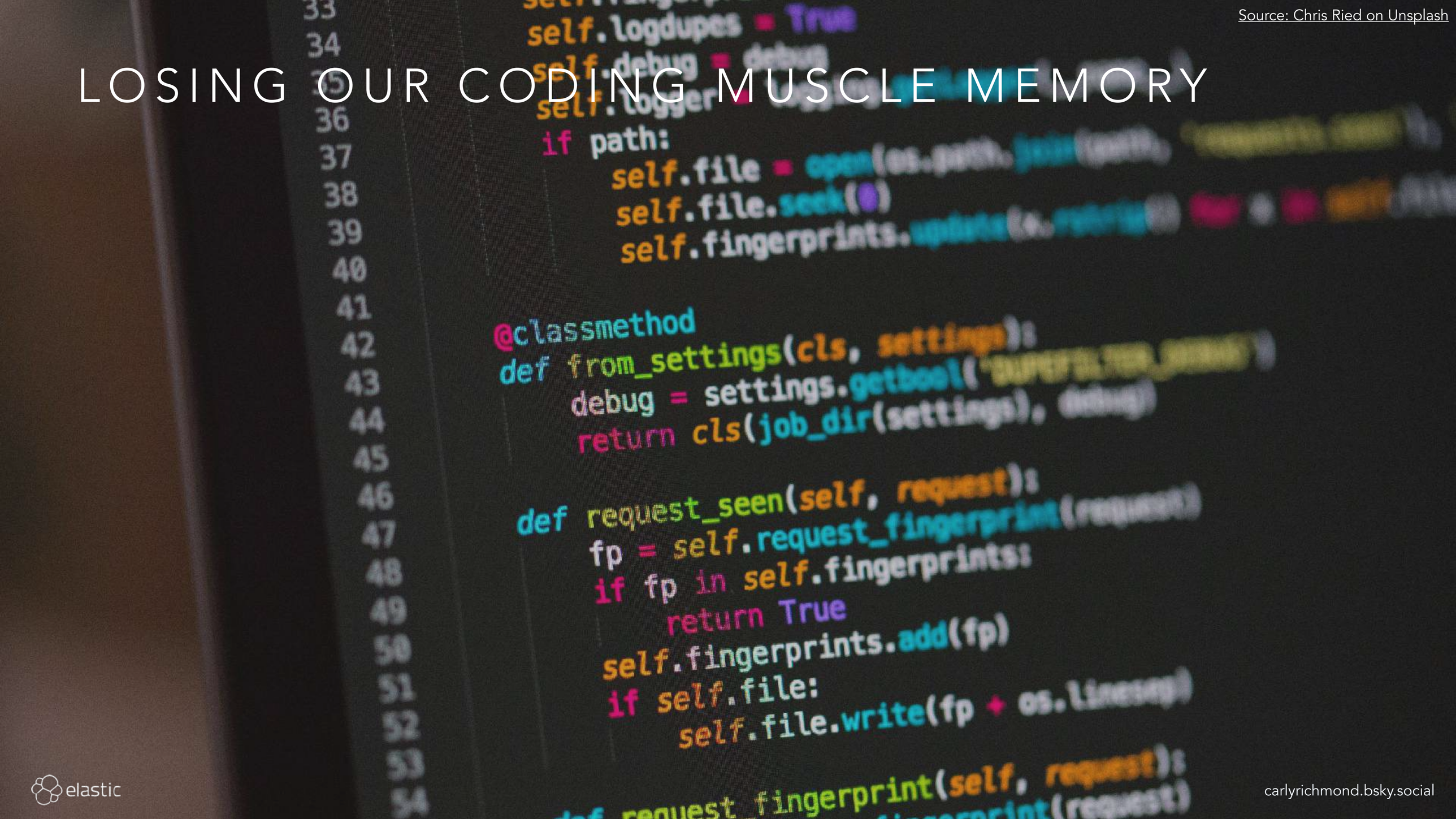
Source: Sasun Bughdaryan on Unsplash

THE LEFTOVER PRINCIPAL



Source: Eiliv Aceron on Unsplash

LOSING OUR CODING MUSCLE MEMORY



Source: Chris Ried on Unsplash

AI ALERT FATIGUE







Source: Various on Unsplash















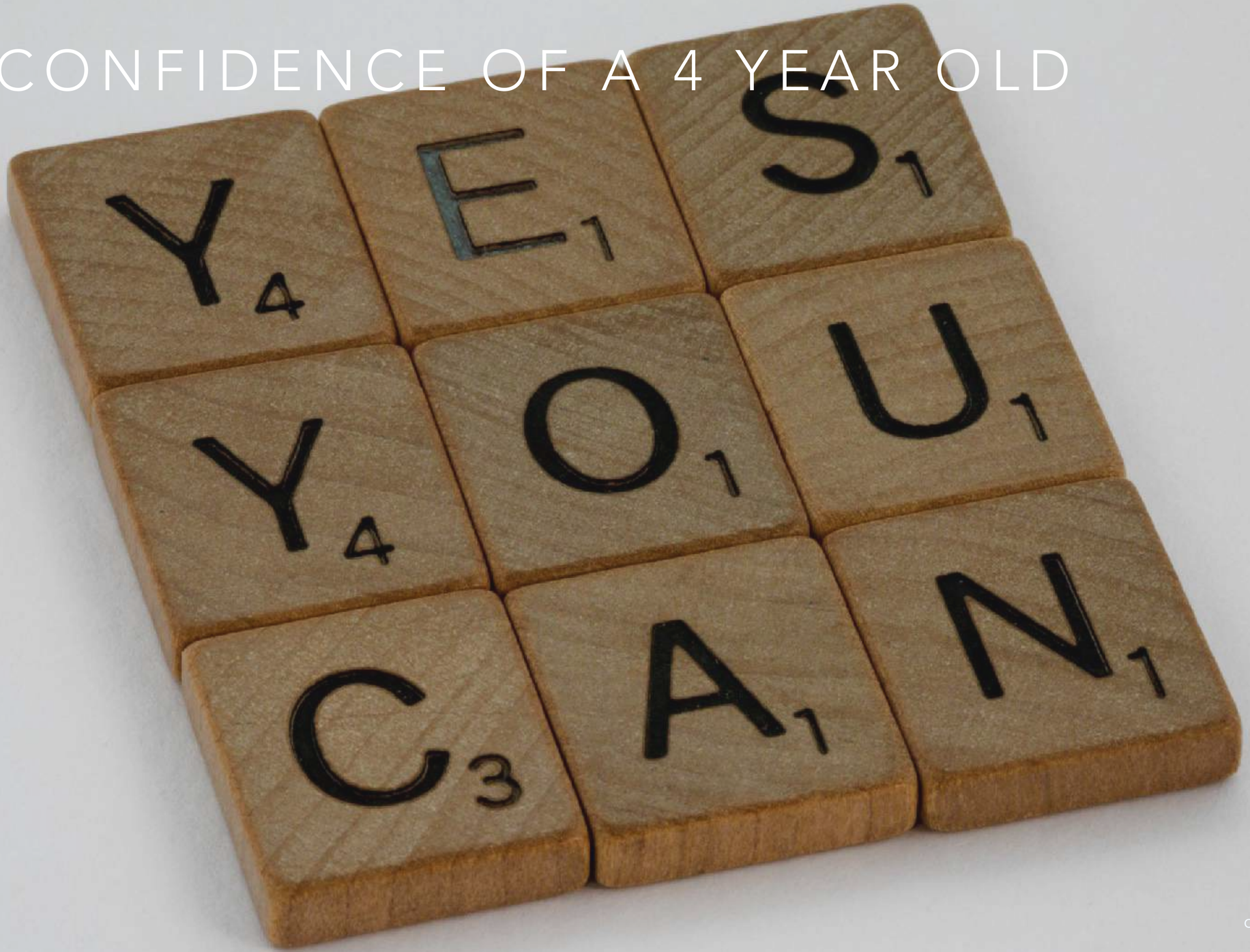




elastic

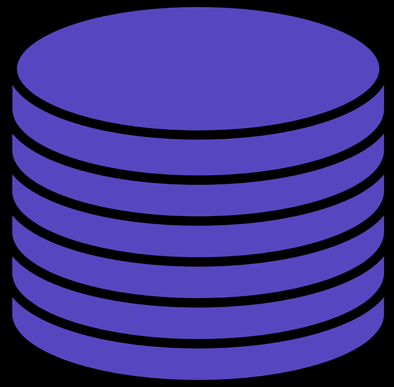
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OVERCONFIDENCE OF A 4 YEAR OLD

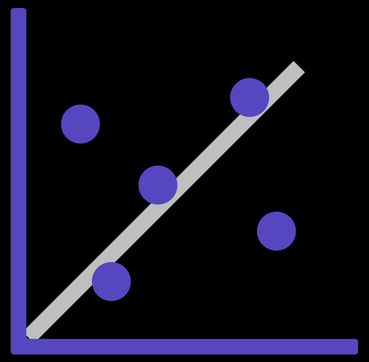


Source: Brett Jordan on Unsplash

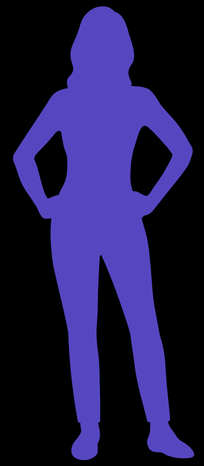
WHY DO LLMS HALLUCINATE?



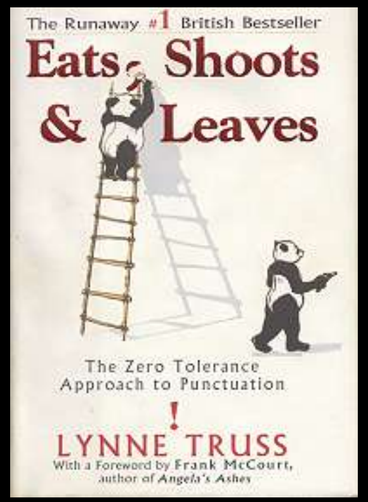
Frozen or Limited Knowledge



Overfitting



Biases



Language Ambiguity



Catastrophic Forgetting

AGENDA

- How incidents have changed
- HITL in incident management
- Benefits of AI
- AI challenges in incident management



Source: Mohammed Sultan Farooqui on Unsplash

RESOURCES

- Electric Youth: Why LLMs are Like My 4-Year-Old
- Growing Pains: Have LLMs aged like a 7-year-old?
- Wargaming in the AI Era

at https://carlyrichmond.com/



Source: Mohammed Sultan Farooqui on Unsplash



Source: Priscilla Du Preez 🇨🇦 on Unsplash